

Rongqian Will Chen

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Education

George Washington University (GW)

PhD in Electrical Engineering

University of Pennsylvania (UPenn)

MSE in Electrical Engineering, GPA: 3.81/4.0

Southwest Jiaotong University (SWJTU)

BEng in Automation, GPA: 3.45/4.0

Washington, D.C., US

Sept. 2024 – Now

Philadelphia, US

Sept. 2021 – May 2023

Chengdu, China

Sept. 2017 – Jun. 2021

Skills

Mathematical Tools Mathematica, Matlab

Design and Simulation SolidWorks, Ansys, Unity, PSIM

Hardware Development Altium Designer, Keil, Proteus, CCStudio

Embedded Systems Linux, Raspberry Pi, Arduino, STM32, FPGA, DSP

Programming Languages Python, C, C++, C#, Verilog, HTML

Experience

Perception Graph Communication Model for MR Systems

Research Leader

Washington, D.C., US

Aug. 2024 – Present

- Developed a Perception Graph model that mimics human interpretation in mixed reality, converting MR visual inputs into a compact semantic graph enriched with multi-functional embeddings.
- Leveraged pre-trained vision–language models to extract and fuse multimodal features, then built a distortion-distance metric to quantify perception attacks in a latent perception space.

AR Guiding System: Assembly King

Lead Developer

Washington, D.C., US

Oct. 2024 – Dec. 2024

- Developed an AR-guided real-time tutoring system on Meta Quest 3 and PC to assist mechanical part assembly, enabling intuitive coach–trainee interactions.
- Engineered computer-vision and Vision-Language Model pipelines to recognize 3D-printed components, infer assembly steps, and provide context-aware highlighting.
- Implemented real-time instruction overlays and feedback loops, streamlining the assembly workflow and reducing errors.

Macro Economics Bot: MarketDigest

Developer

Online

Apr. 2024 – Sept. 2024

- Automated scraping and parsing of macroeconomic reports from 14 major institutions (e.g., BlackRock, Goldman Sachs, IMF, ECB, Fed).
- Designed a RAG pipeline: PDF content extraction, LLM-based summarization, and Pinecone vector database for semantic search.
- Built an interactive chatbot interface supporting natural-language queries on market trends, economic indicators, and institutional outlooks.
- Deployed the system using Docker on AWS ECS with S3 for storage; open-sourced code on GitHub.

Pneumatic Legged Hopping Robot (Master's Thesis, Sung Robotic Lab & Kodlab at UPenn)

Leader

Philadelphia, US

Sept. 2022 – July 2024

- Modeled, fabricated, and characterized a tunable-stiffness pneumatic bellows actuator without external pressure, achieving a 143% stiffness adjustment range via analytical, data-driven, and FEA methods.
- Designed and built a legged hopper robot's mechanical and electrical systems, integrating Raspberry Pi, Pico, Arduino, ESP32 controllers, and a brushless motor controller, plus communication, perception, and data-processing modules.
- Developed a five-bar-linkage ground emulator and its stiffness/damping controller.
- Simulated and validated ground-adaptation energy-saving strategies, demonstrating a 29.3% reduction in energy loss by tuning leg stiffness.

Tunable-Stiffness Coiled-Spring Actuator

Core Member

Philadelphia, US

Oct. 2021 – Sept. 2022

- Developed mathematical models for elastic ring deformation and coil stiffness, simulating multilayer spring force–displacement curves.

- Designed and optimized 3D-printed mechanical structures, transmission mechanisms, and electronic circuits; programmed the coil-spring motor controller.
- Applied the actuator to a flexible manipulator, measured workspace with a motion-tracking system, and implemented an end-effector path-tracking controller.

Moving Target Tracking Algorithm of mmWave Radar

Leader

Chengdu, China

Nov. 2020 – Jul. 2021

- Deployed and calibrated a 77 GHz FMCW radar data-collection system through road testing.
- Developed and optimized signal-processing algorithms to enhance image resolution and reduce detection errors.
- Designed a robust moving target tracking algorithm using limited radar features.

Pulse Power Converter with Dual-Inductance Active Storage Unit

Core Member

Chengdu, China

Sept. 2019 – Aug. 2020

- Designed control circuits using op-amps and improved hysteresis current-control strategies.
- Reduced current spikes and bus voltage ripples through hardware refinements, improving pulse current compensation and lowering power loss.
- Simulated the system in PSIM and validated theory with an experimental prototype.

Acquiring and Stacking Blocks Solution for Robotic Arm

Core Member

Philadelphia, US

Mar. 2022 – May 2022

- Calculated optimal stacking trajectories and optimized algorithms to accelerate robotic arm computations.
- Developed pick-up strategies to maximize stacking height and minimize cycle time under variable conditions.

FPGA-Based Acceleration of Deep Convolutional Neural Networks

Leader

Philadelphia, US

Nov. 2021 – Dec. 2021

- Transformed convolution operations into parallel matrix multiplications, boosting performance and memory bandwidth.
- Deployed PyTorch models on FPGA, developing custom layers in C and Python.

Embedded AI on Raspberry Pi

Member

Online

Jul. 2020 – Oct. 2020

- Built and trained lightweight image-classification models on Raspberry Pi using transfer learning, improving inference accuracy.

Publications

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- **Rongqian Chen**, Jun Kwon, Kefan Wu, Wei-Hsi Chen. Tunable Stiffness for Energy-Efficient Vertical Hopping in a Monopedal Robot Across Varying Ground Profiles. *ICRA 2025*.
 - **Rongqian Chen**, Jun Kwon, Wei-Hsi Chen, Cynthia Sung. Design and Characterization of a Pneumatic Tunable-Stiffness Bellows Actuator. *RoboSoft 2024*.
 - Shivangi Misra, Mason Mitchell, **Rongqian Chen**, Cynthia Sung. Design and Control of a Tunable-Stiffness Coiled-Spring Actuator. *ICRA 2023*.
 - **Rongqian Chen**, Yingquan Zou, Anyong Gao, Leshi Chen. A Cluster-Based Weighted Feature Similarity Moving Target Tracking Algorithm for Automotive FMCW Radar. *VTC 2022-Spring*.
 - Ping Yang, Xi Chen, **Rongqian Chen**, et al. Stability improvement of Pulse Power Supply with dual-inductance Active Storage Unit Using Hysteresis Current Control. *IEEE JETCAS*, 2021.